

5.2.3 Redox and electrode potentials

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
Redox	
(a) explanation and use of the terms <i>oxidising agent</i> and <i>reducing agent</i> (see also 2.1.5 Redox)	
(b) construction of redox equations using half-equations and oxidation numbers	M0.2
(c) interpretation and prediction of reactions involving electron transfer	
Redox titrations	
(d) the techniques and procedures used when carrying out redox titrations including those involving $\text{Fe}^{2+}/\text{MnO}_4^-$ and $\text{I}_2/\text{S}_2\text{O}_3^{2-}$ (see also 2.1.5 e–f)	HSW4 Opportunities to carry out experimental and investigative work.
(e) structured and non-structured titration calculations, based on experimental results of redox titrations involving: (i) $\text{Fe}^{2+}/\text{MnO}_4^-$ and $\text{I}_2/\text{S}_2\text{O}_3^{2-}$ (ii) non-familiar redox systems	M0.1, M0.2, M0.4, M1.1, M1.2, M2.2, M2.3, M2.4 Non-structured titration calculations could be examined in the context of both acid–base and redox titrations (see also 2.1.4 d–e).
Electrode potentials	
(f) use of the term <i>standard electrode (redox) potential</i> , $E^\ominus$, including its measurement using a hydrogen electrode	$E^\ominus$ data will be provided on examination papers.
(g) the techniques and procedures used for the measurement of cell potentials of: (i) metals or non-metals in contact with their ions in aqueous solution (ii) ions of the same element in different oxidation states in contact with a Pt electrode	For measurement of standard cell potentials, ions of the same element can have concentrations of 1 mol dm^{-3} or be equimolar. PAG8 HSW4 Opportunities to carry out experimental and investigative work.
(h) calculation of a standard cell potential by combining two standard electrode potentials	
(i) prediction of the feasibility of a reaction using standard cell potentials and the limitations of such predictions in terms of kinetics and concentration	M0.3 HSW6 The relative effects of standard electrode potential, rate and concentration in determining feasibility of processes.

Storage and fuel cells

- (j) application of principles of electrode potentials to modern storage cells

Details of storage cells and required equations will be provided. Relevant electrode potentials and other data will be supplied.

HSW9 Benefits of electrochemical cells counteracted by risks from toxicity and fire from Li-based cells.

- (k) explanation that a fuel cell uses the energy from the reaction of a fuel with oxygen to create a voltage and the changes that take place at each electrode.

Recall of fuel cells and equations will **not** be required. Relevant electrode potentials and other data will be supplied.